

Thermodynamic performance of Kalina cycle system 11 (KCS11): feasibility of using alternative zeotropic mixtures

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Abstract

With the ever increasing demand for energy, exploiting low-temperature heat sources has seen significant interest recently. The conventional organic Rankine cycle (ORC) is a typical approach used to exploit low-temperature heat sources but suffers from low efficiency. The Kalina cycle is a reversed absorption cooling system that normally utilizes an ammonia–water binary mixture as the working fluid. This paper investigates, using thermodynamic modelling, the performance of Kalina cycle system 11 (KCS11) used for low-temperature heat sources below 200°C compared with the ORC based on pure ammonia and R134a. The cycle performance was investigated at various operating conditions including the evaporator pressure of 10–50 bars, heat source temperature of 333–473 K, heat sink temperature of 283 K and in the case of KCS11 various ammonia mass fractions at the evaporator outlet. Results show that the KCS11 can produce up to 40% increase in the efficiency compared with the ORC when using ammonia and up to 20% increase when using R134a. Although the ammonia–water working pair has zero ozone depletion potential (ODP) and very low global warming potential (GWP), it is toxic and needs special safety procedures against leak as ammonia is part of this binary mixture. Therefore, further investigation was carried out to explore the feasibility of using alternative working pairs that are non-toxic and outperform the ammonia–water pair for the Kalina cycle. Nineteen working pairs were investigated and results showed that propane and propylene-based mixtures have the potential to replace the ammonia–water pair in the KCS11.

Keywords: Kalina cycle (KCS11); ORC; ammonia; R134a; propane; propylene

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1 INTRODUCTION

With the increasing demand for and cost of energy, the exploitation of low-grade heat sources such as geothermal, solar and waste heat sources has been receiving more attention. Through the advancement of technology, there is much interest in designing more efficient, reliable and cost-effective energy conversion systems that will provide means of exploiting low-temperature heat sources that might not otherwise be utilized. The Kalina cycle and the organic Rankine cycle (ORC) provide possible solutions to the problem of recovering the low-temperature energy that is usually thrown away as waste heat; with the ORC has a drawback of low overall efficiency [1]. Interest in the Kalina cycle has been growing, since it was patented by Dr Alexander Kalina in the 1980s. The Kalina cycle is a modified conventional ORC or reversed absorption

cycle [2], and it is the first major advancement in the power generation technology over the Rankine cycle that was invented by William Rankine of Scotland over 150 years ago. Compared with conventional thermodynamic cycles, the Kalina cycle power plant can offer an efficiency gain of 10–50% for low-temperature thermal energy sources such as geothermal brine at 60–200°C [3], waste heat from gas turbines [4, 5] and waste heat from iron and steel industry. It is likely that Kalina cycle plants can cost even less to build than Rankine cycle plants with an equal output. Up to 30% savings for applications with low-temperature heat sources and up to 10% savings for direct fired or bottoming cycle plants have been reported by Global Geothermal Limited [3].

Generally, there are different types of Kalina cycle families, which are known by their unique names. For example, KCS5 is particularly applicable to direct fired plants. KCS6 is applicable

Table 1. Investigated working pairs for the KCS11

Proposed binary mixtures						
CO ₂ mixtures	R32 mixtures		Propane mixtures		Propylene mixtures	
CO ₂ –DME	R32–DME		R290–R601		R1270–R601	
CO ₂ –R1270	R32–R600		R290–R600		R1270–R600	
CO ₂ –R290	R32–R600a		R290–R600a		R1270–R600a	
CO ₂ –R601a	R32–R601a		R290–R601a		R1270–R601a	
CO ₂ –R601						
CO ₂ –R600a						
CO ₂ –R600						
Refrigerant	NPB (°C)	GWP	Hfg (kJ/kg)	Flammability	Toxicity	ASHRAE safety [14]
Ammonia (R717)	–33.34	<1	1370	Yes	Yes	B2
Water (R718)	100	0	2256	No	No	A1
Carbon Dioxide (R744)	–78.46	1	232	No	No	A1
Difluoromethane (R32)	–51.65	650	382	Yes	No	A2
Propylene (R1270)	–47.62	3	438	Yes	No	A3
Propane (R290)	–42.11	3	425	Yes	No	A3
Butane (R600)	–0.49	3	386	Yes	No	A3
Iso-butane (R600a)	–11.749	3	365	Yes	No	A3
Pentane (R601)	36.06	3	357	Yes	No	A3
Iso-pentane (R601a)	27.5	3	343	Yes	No	A3
Dimethylether (DME)	–24.782	2	465	Yes	No	A3

to gas turbine based on combined cycles and Kalina cycle system 11 (KCS11) and KCS34 are designed for exploiting low-temperature heat sources. For this work, KCS11 was selected as it is the most applicable for low-grade waste heat sources at temperatures below 200°C [6]. In this paper, the thermodynamic analysis of the KCS11 using ammonia–water was compared with that of the ORC based on pure ammonia or pure R134a at various operating conditions.

Although the ammonia–water working pair has zero ozone depletion potential (ODP) and very low global warming potential (GWP), it is toxic and needs special safety procedures against leak. Therefore, there is a need to explore the feasibility of using other working pairs to replace water–ammonia in the KCS11. Recently, extensive research was carried out for developing mixed refrigerants in the field of refrigeration and air conditioning including mixing of CFC (chlorofluorocarbon), HCFC (hydrochlorofluorocarbons), HFC (hydrofluorocarbons) and commercial products of such blends are available such as R407C. Also some mixtures have been reported in literature including CO₂–hydrocarbon blends [7], CO₂–dimethylether (DME) [8] and R32–hydrocarbons [9]. The choice of these refrigerants is based on their environmental favourable characteristics such as zero ozone depletion, low GWP and non-toxicity. Mixing hydrocarbon refrigerants with CO₂ reduce their flammability and offer good control on the pressure level of carbon dioxide based on the mixing concentration. Additionally, R32 is an energy efficient refrigerant due to its relatively high pressure and density; as a result, R32 mixtures could be comparable with ammonia–water mixtures. It has been reported that zeotropic mixtures of HFC could be used in the Kalina cycle such as R22–R134a by Shin *et al.* [10] and R32–R134a by Kim *et al.* [11]. The principle of forming

the zeotropic mixture is to mix fluids with different boiling points so that the evaporation or condensation process occurs over a temperature range (temperature glide).

In this work, 19 working pairs were investigated for replacing the water–ammonia working pair in the KCS11 as shown in Table 1. These mixtures are classified in four groups based on the component with low boiling point, namely, CO₂, R32, propane and propylene.

2 KCS11 AND ORC THERMAL MODELLING

Figure 1a shows a schematic diagram of the KCS11. It consists of a turbine, absorber, condenser, evaporator, separator, regenerator, pump and throttling valve. In the evaporator, the ammonia–water mixture is heated up by the low-temperature heat source and then passes to the separator. In the separator, the saturated vapour part of the mixture separates from the liquid. The rich ammonia saturated vapour mixture then expands through the turbine producing power output and then it passes through the absorber. The ammonia–water solution leaves the absorber to the condenser where it is condensed and then it is pumped to increase its pressure to that corresponding to the evaporator. The hot weak ammonia–water saturated liquid mixture leaving the separator is then directed to the regenerator where it is cooled by the rich ammonia mixture flowing back to the evaporator. After the regenerator, the weak ammonia solution passes through a throttling valve to lower its pressure. The ORC consists of four components, namely the turbine, evaporator, condenser and pump, as shown in Figure 1b. In the ORC modelled in this paper, pure ammonia or R134a were used as the working fluid.

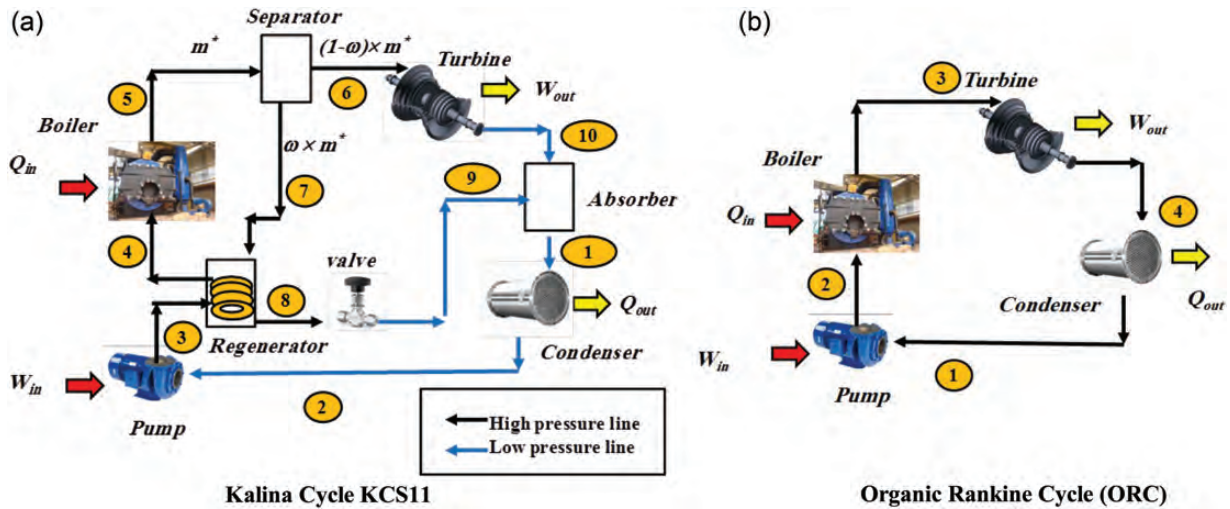


Figure 1. Flow diagram of different cycles: (a) KCS11 and (b) ORC.

The modelling of the KCS11 is carried out by applying the steady flow energy and mass balance equations to the various components of the system neglecting changes to kinetic and potential energy and frictional losses. Assuming that both the pump (η_{pump}) and turbine (η_{turb}) have isentropic efficiency of 80%, the specific work required by the pump (w_{pump}) and the specific work produced by the turbine (w_{turb}) were calculated by:

$$w_{\text{pump}} = \left(\frac{v_2 \times (P_3 - P_2)}{\eta_{\text{pump}}} \right) = h_3 - h_2, \quad (1)$$

$$w_{\text{turb}} = \left((1 - \omega) \times \frac{h_6 - h_{10,s}}{\eta_{\text{turb}}} \right) = (1 - \omega) \times (h_6 - h_{10}), \quad (2)$$

where ω is the ratio of the mass flow rate of the ammonia weak solution leaving the separator to the regenerator (state 7) and the mass flow rate of the ammonia-rich solution entering the separator (state 5). v_2 , h_6 and h_{10} are the specific volume at the pump inlet, specific enthalpy at the turbine inlet and specific enthalpy at the turbine outlet obtained as the function of temperature, pressure and ammonia concentration in the solution. $h_{10,s}$ is the specific enthalpy of the ammonia–water solution assuming isentropic expansion through the turbine. In all the modelling, the dryness fraction at the exit of the turbine was maintained higher than 90% to minimise the liquid droplet formation in the turbine.

The pressure reducing valve after the regenerator is assumed to be adiabatic, thus the enthalpy of the fluid at the inlet is equal to that at the exit of the valve:

$$h_9 = h_8. \quad (3)$$

The separator and the absorber are assumed to be adiabatic with no external heating or cooling applied:

$$h_5 - h_6 = \omega \times (h_7 - h_6), \quad (4)$$

$$h_1 - h_{10} = \omega \times (h_9 - h_{10}). \quad (5)$$

For the regenerator, assuming no heat losses to the surrounding and a minimum temperature difference (Pinch point) of 4 K, the rate of energy absorbed by the ammonia-rich solution (state 3 to state 4) is equal to the heat lost by the ammonia weak solution (state 7 to state 8) thus:

$$h_4 - h_3 = \omega \times (h_7 - h_8). \quad (6)$$

For the evaporator and condenser, the specific energy absorbed from the heat source and that rejected to the heat sink are given by:

$$q_e = h_5 - h_4, \quad (7)$$

$$q_c = h_1 - h_2. \quad (8)$$

The thermal efficiency of the KCS11 can then be determined from:

$$\eta = \left(\frac{W_{\text{net}}}{q_e} \right) \times 100, \quad (9)$$

where the net power output is determined by:

$$W_{\text{net}} = (1 - \omega) (h_6 - h_{10}) - (h_3 - h_2). \quad (10)$$

The modelling of the ORC has been carried out in order to compare the KCS11 performance to that of the ORC. Applying the steady flow energy equation on the components of the ORC, the net power output, the evaporator heat input and the cycle thermal efficiency can be determined using Equations (11)–(15) with the minimum temperature difference between the working fluid and the heat source in the evaporator or heat

sink in the condenser being 4 K:

$$w_{\text{pump}} = \frac{v \times (P_2 - P_1)}{\eta_{\text{pump}}} = h_2 - h_1, \quad (11)$$

$$w_{\text{turb}} = \frac{h_3 - h_{4,s}}{h_{\text{turb}}} = h_3 - h_4, \quad (12)$$

$$w_{\text{net}} = (h_3 - h_4) - (h_2 - h_1), \quad (13)$$

$$q_e = h_3 - h_2, \quad (14)$$

$$\eta = \frac{w_{\text{net}}}{q_e} \times 100. \quad (15)$$

The modelling was conducted using Engineering Equations solver (EES) where pure ammonia and pure R134a thermodynamic properties are available. Also, the ammonia–water mixture properties are based on a formulation by Ibrahim and Klein [12]. For the 19 working pairs listed in Table 1, the Refprop software was linked to the EES to carry out the simulation.

3 COMPARING AMMONIA–WATER KCS11 TO ORC

This work investigated the performance of the KCS11 compared with the ORC in terms of its effectiveness in all

applications that produce heat at temperatures of $<200^\circ\text{C}$. The ORC utilized either pure ammonia or pure R134a as a working fluid, while the KCS11 utilised the ammonia–water mixture. Figure 2a–c shows the thermal efficiency curves for the KCS11 as a function of the ammonia mass fraction at the evaporator outlet for several heat source temperatures. In these graphs, the heat sink temperature was set at 283 K and the heat source temperature varied from 333 K (Figure 2a), 373 K (Figure 2b), to 423 K (Figure 2c). It should be mentioned that using the ammonia–water mixture at more than 400°C is not advisable, because at higher temperature NH_3 becomes unstable which leads to nitride corrosion [13]. The results show that as the heat source temperature increases, the maximum thermal efficiency of the Kalina cycle increases. Also, the results show that when the ammonia concentration is too lean in the working fluid; the thermal efficiency of the cycle drops quickly. This trend can be explained as follows. At specific temperature and pressure, as the concentration of ammonia decreases, the mixture leaving the evaporator becomes saturated or even sub-cooled liquid. Thus, there will be little or no vapour generated during the separation process; hence, the turbine work output becomes negligible and the efficiency reduces steeply. On the other hand as the ammonia mass fraction increases, the thermal efficiency of the cycle drops gradually. This indicates that, for a KCS11 in operation, the ammonia mass fraction of the working fluid need to be rich to avoid an entire loss in the thermal efficiency of the cycle. Thus, to keep the cycle at a

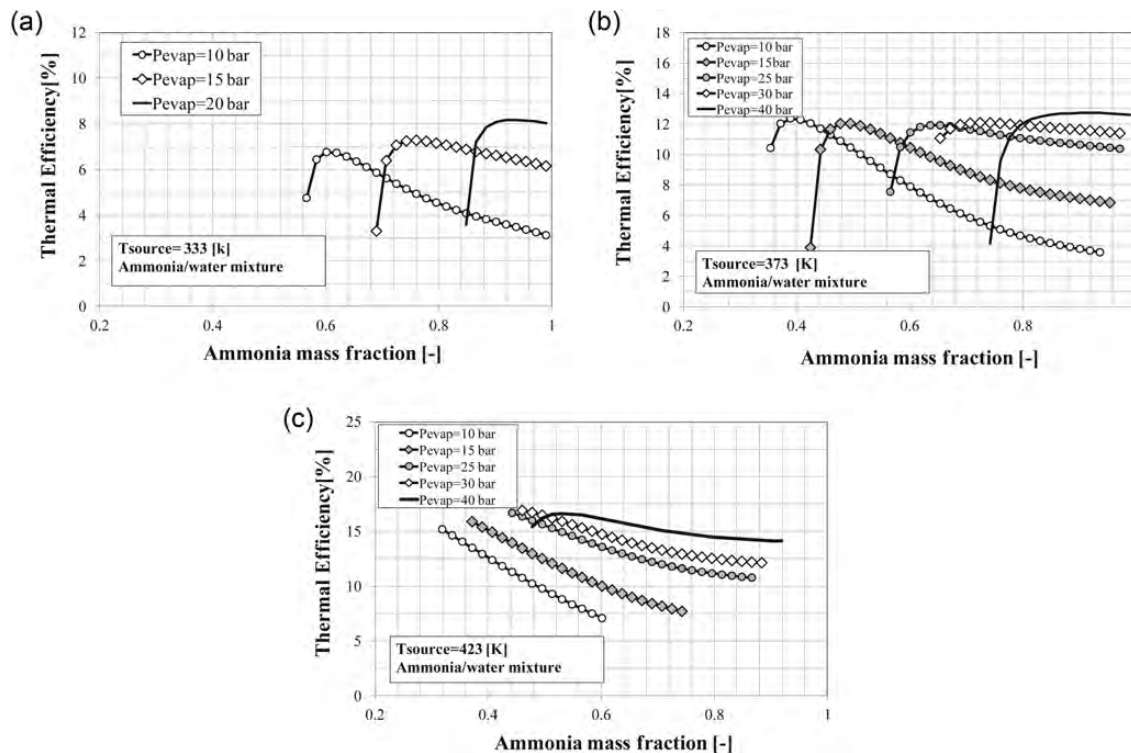


Figure 2. (a) Thermal efficiency of the KCS11 with a source temperature of 333 K and a sink temperature of 283 K. (b) Thermal efficiency of the KCS11 with a source temperature of 373 K and a sink temperature of 283 K. (c) Thermal efficiency of the KCS11 with a source temperature of 423 K and a sink temperature of 283 K.

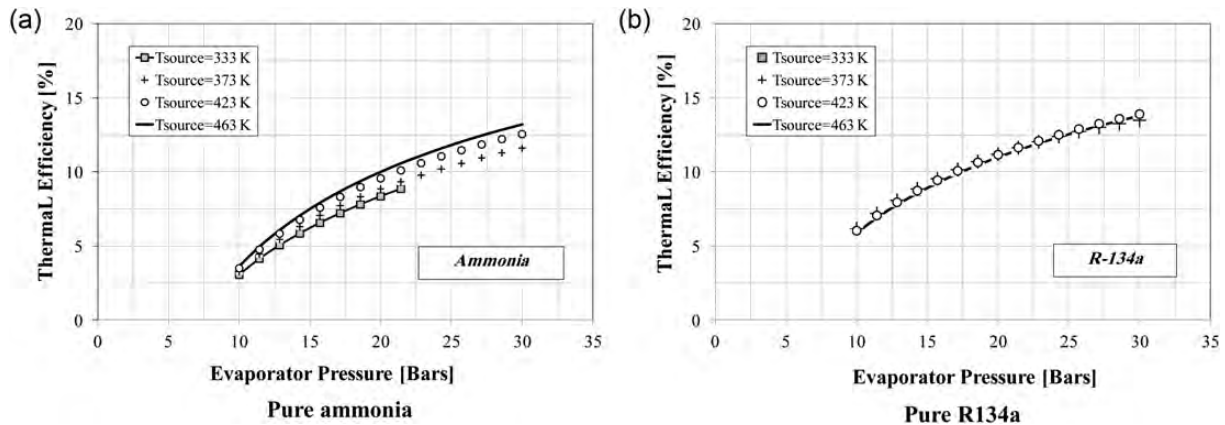


Figure 3. ORC thermal efficiency at various evaporator pressures and temperatures at a heat sink temperature of 283 K: (a) pure ammonia and (b) pure R134a.

reasonable efficiency and stable operating conditions, the mass fraction of ammonia should be in the range of 0.55–0.9. The figures also show that for fixed evaporator pressure, the point of maximum efficiency shifts towards low-concentration values by increasing the evaporator temperature (heat source).

Figure 3 shows the thermal efficiency of the ORC using pure ammonia (Figure 3a) and pure R134a (Figure 3b) as the working fluid. To compute the thermal efficiency of the cycle, the evaporator pressure was increased while holding the heat sink and heat source temperatures constant. The evaporator pressure was limited so that the turbine exit quality was no less than 90%. It is clear from this figure that, as the heat source temperature and evaporator pressure increases, the thermal efficiency of the ORC increases. However, the effect of the heat source temperature is more noticeable in the case of ammonia compared with that of R134a. Figure 3 also shows that the maximum efficiency obtained was 14% for R134a and 13% for ammonia at heat source temperature of 463 K and evaporator pressure of 30 bars.

Figure 4 compares the Kalina cycle to the ORC using ammonia and R134a in terms of the thermal efficiency at heat source temperature of 373 K. Two ammonia–water concentration values were used, 0.66 and 0.55. It can be seen that the thermal efficiency of the Kalina cycle with the ammonia–water concentration of 0.55 is significantly higher than that of the ORC using ammonia and R134a at evaporator pressures below 20 bars. For example, at a pressure of 15 bars, the KCS11 thermal efficiency (11.38%) with the ammonia–water concentration of 0.55 is ~40% higher than that of the ORC using pure ammonia (7%) and 20% higher than that of the ORC using pure R134a (9.2%) with the heat source temperature of 373 K and the heat sink temperature of 283 K. This improvement in the KCS11 efficiency compared with the ORC is mainly due the variable boiling and condensation temperatures of the binary mixture which provides better match to the heat

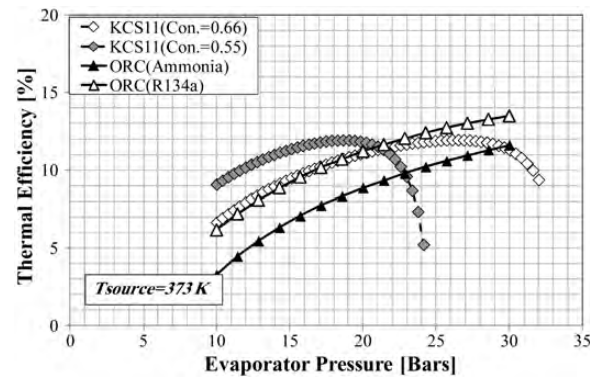


Figure 4. Comparison between the Kalina cycle and the ORC based on pure ammonia and pure R134a at an evaporator temperature of 373 K and a heat sink temperature of 283 K.

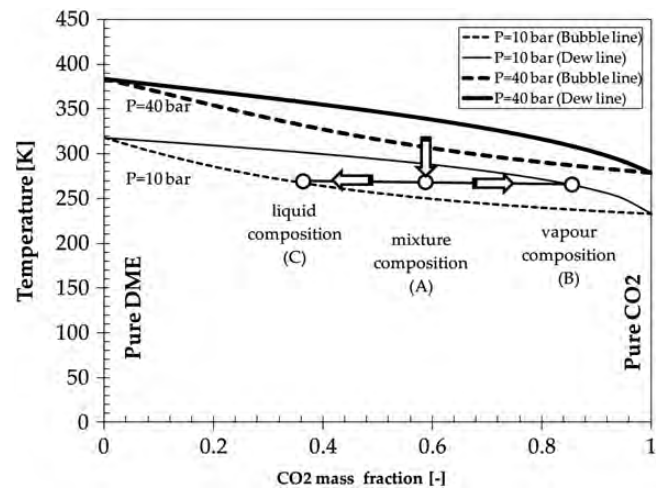


Figure 5. Phase equilibrium diagram for the zeotropic refrigerant mixture CO_2 –DME.

source and heat sink temperatures with lower temperature differences and reduced thermal irreversibility. The ideal Carnot cycle efficiency for the heat source temperature of 373 K and the heat sink temperature of 283 K is 24%; therefore, the

second law efficiency (the ratio of cycle efficiency to that of the Carnot cycle) for this reported KCS11 is 47% which highlights the potential of this cycle. At evaporator pressures higher than 20 bars, the KCS11 thermal efficiency decreases significantly to

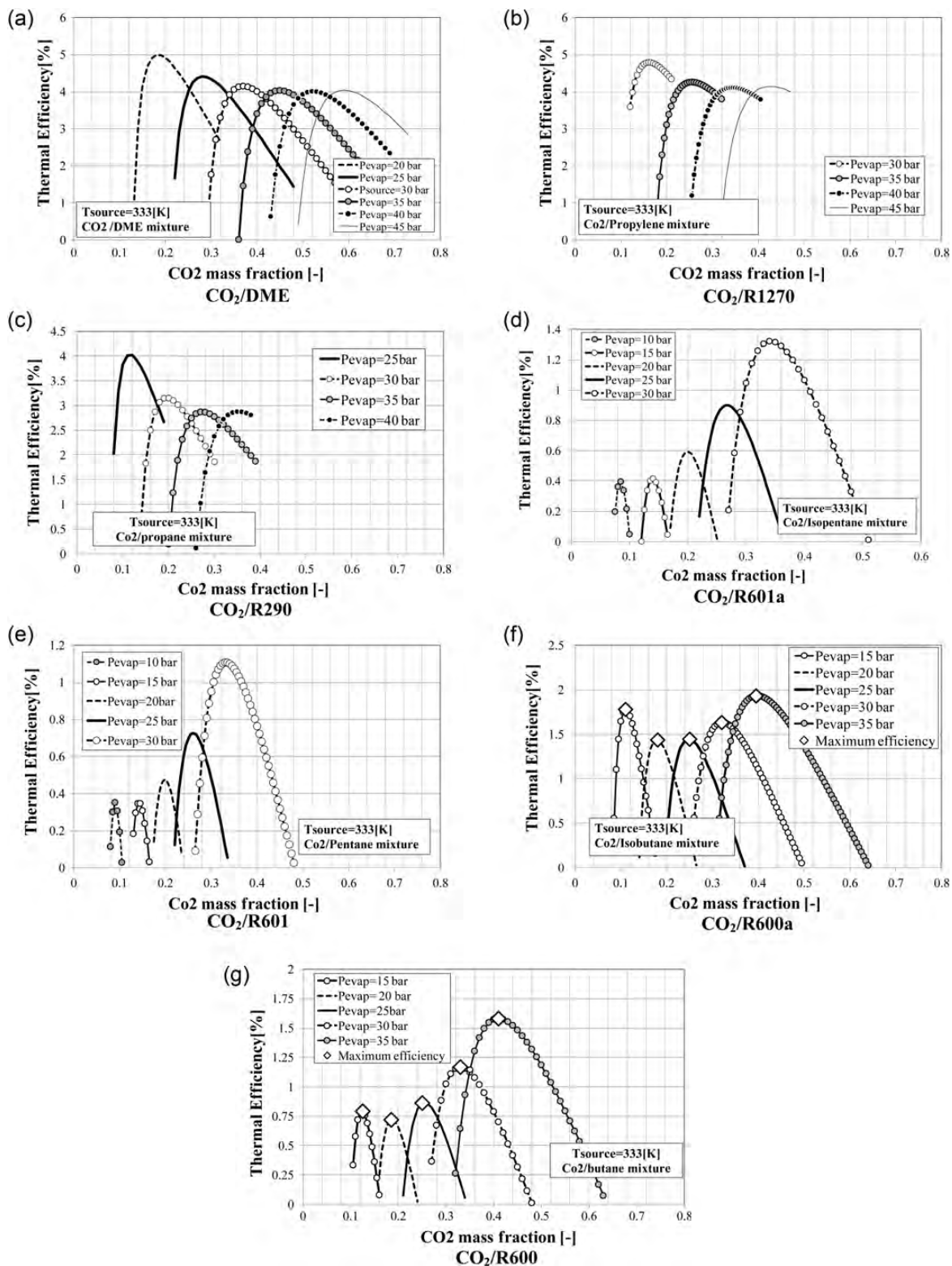


Figure 6. Carbon dioxide refrigerant mixtures ($T_{source} = 333\text{ K}$, $T_{sink} = 283\text{ K}$).

be lower than those of the ORC. For the KCS11 with the ammonia–water concentration of 0.66, its thermal efficiency is consistently higher than those of the ORC using ammonia over the wide range of evaporator pressure values used but with similar values to those of the ORC using R134a. The high efficiency of the KCS11 at low operating pressures results in economic advantages in terms of the lower system cost.

4 ALTERNATIVE FLUIDS FOR KCS11

In this section, 19 working pairs were investigated for replacing the water–ammonia working pair in the KCS11 as shown in Table 1. The Refprop thermophysical platform was linked to the EES software where the Kalina cycle code was executed. Figure 5 presents the equilibrium phase diagram of carbon dioxide–dimethyl ether blends at 10 and 40 bars as generated by the Refprop package. The dew line represents the saturated vapour line and the bubble line represents the saturated liquid line. The axis on left-hand side represents pure dimethyl ether with the higher saturation temperature (317 K at 10 bars) and the axis on the right-hand side represents pure carbon dioxide with the lower saturation temperature (233 K at 10 bars). It should be mentioned that the flow leaving the evaporator of the Kalina cycle should be in the two-phase region (point A), while after the separator, the mixture is separated into vapour and liquid with composition B and composition C,

respectively. This figure was used to calculate the composition of the working fluid in both liquid and vapour phases after the separation process. Also, this figure was used to determine the range of source temperatures that can be used at a specific operating pressure. Similar equilibrium phase diagrams were used for the other blends shown in Table 1.

Figure 6 presents the KCS11 thermal efficiency versus mass fraction for the seven carbon dioxide blends shown in Table 1 for the heat source temperature of 333 K and the heat sink temperature of 283 K. Results show that the performance of CO₂–DME and CO₂–R1270 are better than the other carbon dioxide mixtures. However, their efficiency is significantly lower than that of the water–ammonia mixture (Figure 2a). Also, the maximum efficiency concave trend with increasing the pressure is clearly observed in the case of CO₂–butane (R600) and CO₂–isobutene (R600a), as shown from the peaks. Figure 7 presents the simulated R32 pairs including R32–DME, R32–R600, R32–R600a and R32–R601a. Results show that for the same evaporator pressure, the maximum efficiency of R32–DME is the highest among all the R32 mixtures. Also, the thermal efficiency of R32–R601a is higher than that of R32–R600 and that of R600a. However, comparing Figure 7d with Figure 2a, it is shown that the efficiency of ammonia–water is higher than that of R32–R601a.

Figures 8 and 9 show the KCS11 thermal efficiency versus mass fraction for propane and propylene-based mixtures. From these figures, it is clear that none of the mixtures investigated

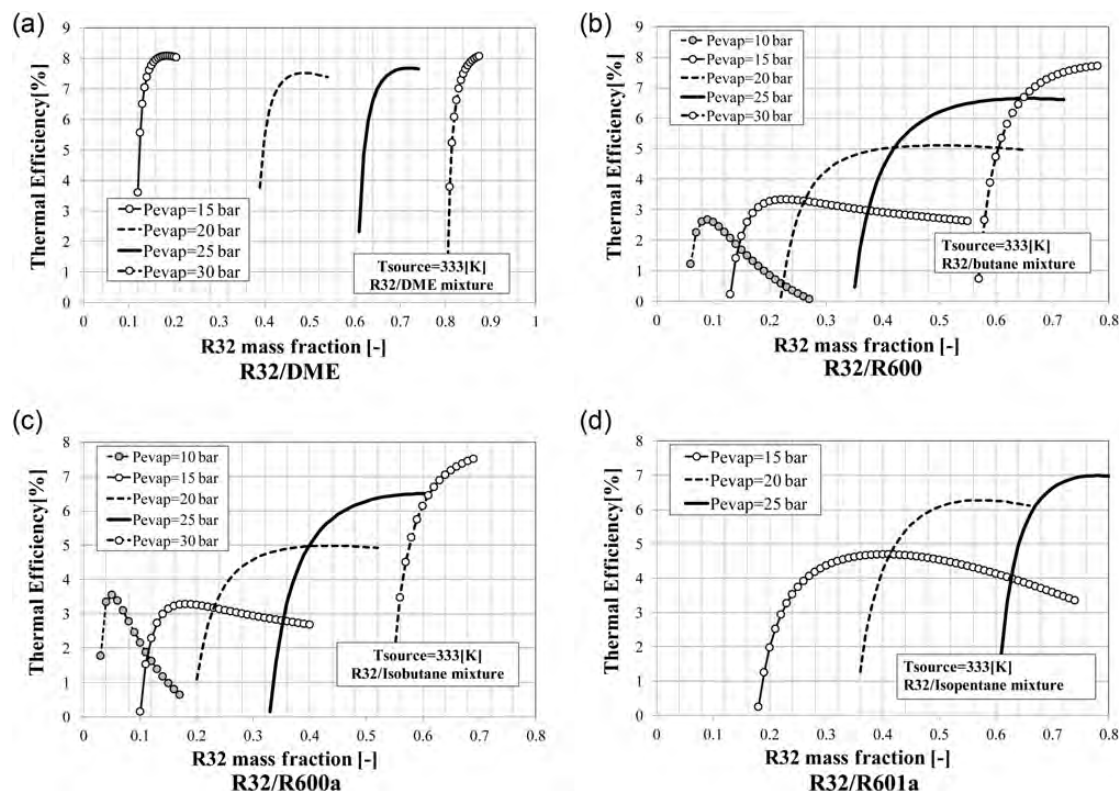


Figure 7. R32 refrigerant mixtures ($T_{\text{source}} = 333 \text{ K}$, $T_{\text{sink}} = 283 \text{ K}$): (a) R32–DME, (b) R32–R600, (c) R32–R600a and (d) R32–R601a.

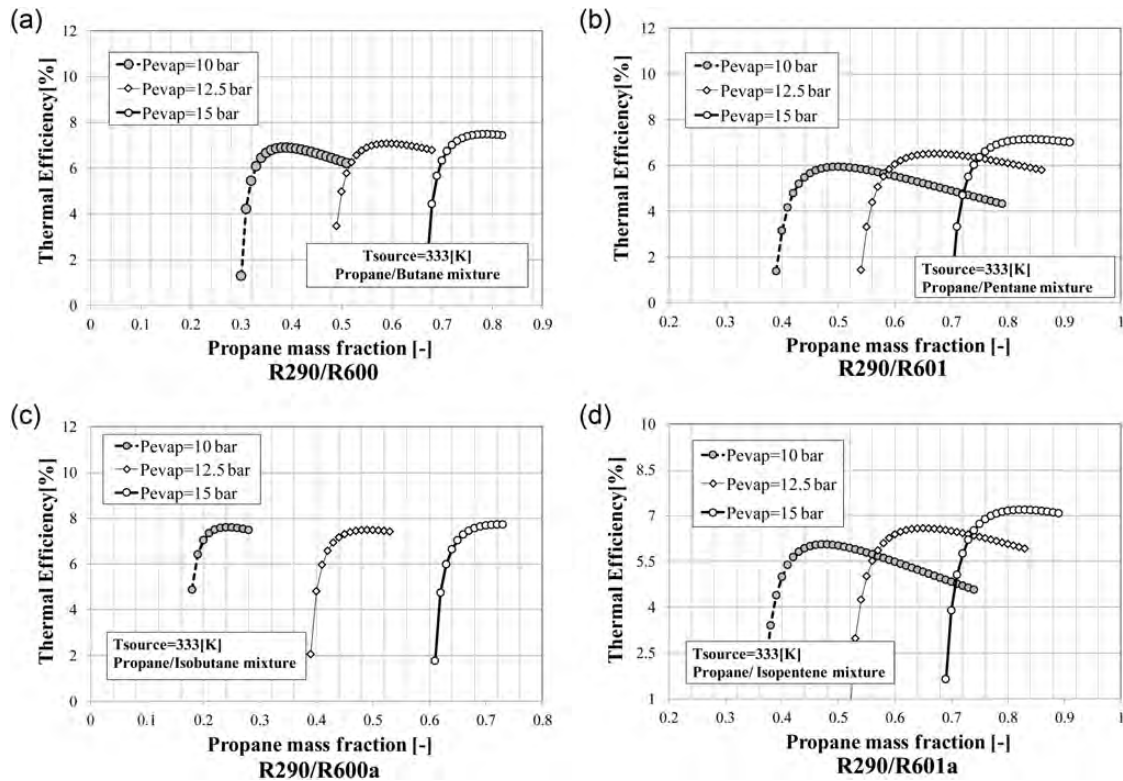


Figure 8. R290 refrigerant mixtures ($T_{source} = 333\text{ K}$, $T_{sink} = 283\text{ K}$): R290–R600, R290–R601, R290–R600a and R290–R601a.

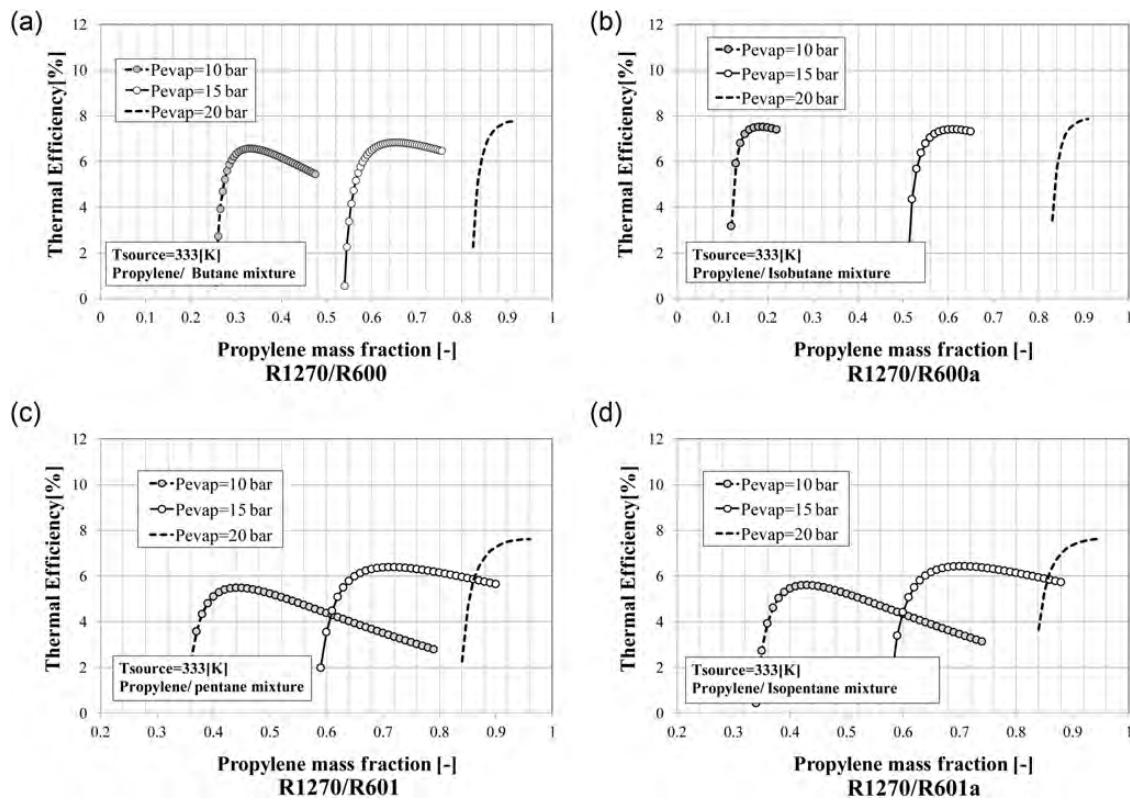


Figure 9. R1270 refrigerant mixtures ($T_{source} = 333\text{ K}$, $T_{sink} = 283\text{ K}$): (a) R1270–R600, (b) R1270–R600a, (c) R1270–R601 and (d) R1270–R601a.

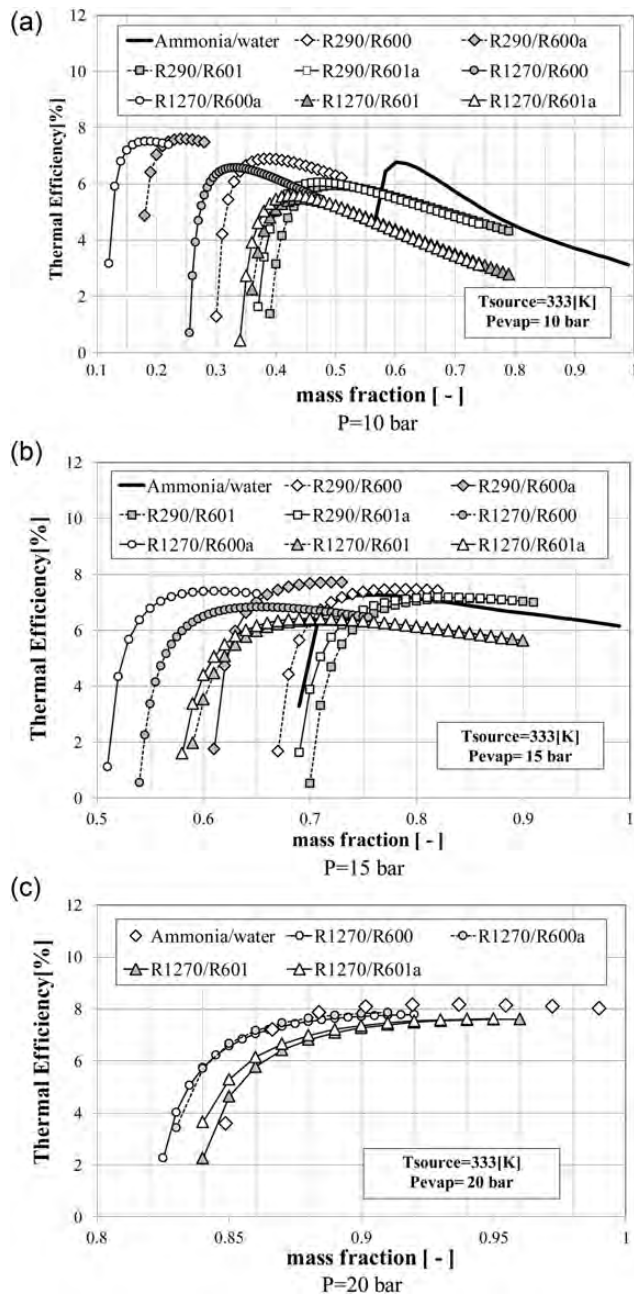


Figure 10. Comparing different refrigerants mixtures ($T_{\text{source}} = 333 \text{ K}$, $P = 15 \text{ bars}$): (a) $P = 10 \text{ bars}$, (b) $P = 15 \text{ bars}$ and (c) $P = 20 \text{ bars}$.

outperform ammonia–water mixture. However, most of these mixtures have comparable performance with that of ammonia–water in 10–20 bars operating pressures. Figure 10 compares the KCS11 performance using the various pairs that were identified to produce thermal efficiency comparable with that of the ammonia water mixture at 333 K heat source temperature and evaporator pressures of 10, 15 and 20 bars. At the 10-bar pressure, Figure 10a shows that R290–R600a and R1270–R600a outperforms the ammonia–water at the mass

fraction of 0.15–0.25 and R290–R600 and R1270–R600 have comparable performance with that of the ammonia–water mixture for the mass fraction of 0.3–0.5. At the 15-bar pressure, Figure 10b shows that R290–R600a, R1270–R600a, R290–R600 and R1270–R600 mixtures have comparable performance with that of the ammonia–water mixture for the mass fraction of 0.55–0.8. At the 20-bar pressure, Figure 10c shows that propylene mixtures have comparable performance with that of the ammonia–water mixture for the mass fraction of 0.85–0.95.

5 CONCLUSIONS

The performance of the KCS11 using the ammonia–water mixture as the working fluid was modelled and compared with the performance of the ORC using pure ammonia or pure R134a as the working fluids. The results reveal that the KCS11 with the ammonia–water concentration of 0.55 achieves 20–40% higher efficiency than the ORC using the same operating conditions of the evaporator pressure of 15 bars, heat source temperature of 373 K and a heat sink temperature of 283 K. The high efficiency of the Kalina cycle at low evaporator pressures will result in reducing the cost of the cycle components thus offsetting the cost of increasing the number of components and can lead to a cost-effective power generation system. The results also show that at a given evaporator pressure, heat source and sink temperatures, an optimum ammonia mass fraction that can give the maximum cycle efficiency can be determined.

Concerns about the toxicity of ammonia led to investigating the potential of other non-toxic working pairs that can either outperform or have a comparable performance with that of ammonia–water. Nineteen mixtures were investigated with results, showing that based on the mass fraction and the evaporator operating pressure, certain propane and propylene-based mixtures can outperform the ammonia–water mixtures, whereas others have similar performance. Such results show the potential of such mixtures and indicate the need for further research.

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